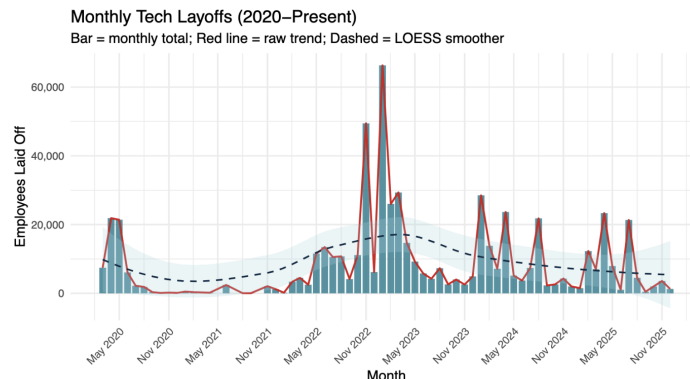


TECH EMPLOYMENT IN THE AGE OF AI: CAN YOU FORECAST WHAT'S NEXT? A DS 4002 Case Study by Jensen Harvey



Tech layoffs dominated the news cycle through 2022 and 2023. Meta cut 11,000. Google cut 12,000. Amazon and Microsoft each cut 10,000 in a single month. By 2025, layoff numbers in the United States rivaled the early COVID era, and for the first time since 2021, more Americans were unemployed than there were open jobs.

Most analysts agree that artificial intelligence is reshaping the tech workforce. What's less clear is what comes next. Will layoffs return to pre-2022 lows? Stabilize at a new normal? Spike again as more companies restructure around AI? These questions matter for workforce policy, for company strategy, and for any UVA student trying to plan a career in technology.

Time series forecasting is one of the standard tools data scientists use to answer questions like these. A previous DS 4002 team built a SARIMA model on Layoffs.fyi data and beat their seasonal naive baseline. They forecast layoffs stabilizing around 8,900 per month through 2027 with recurring Q1 spikes. But they were the first to flag the limitations of their work: public reporting bias in the data, no macroeconomic covariates, only eight months of test data, no industry-level breakdowns.

You are a data scientist stepping into the next phase of this project. Your task is to choose a meaningful extension of the original work, justify your direction, and produce a forecast and writeup that improves on what came before. You will work with real public data and real economic uncertainty, and you will wrestle with the same questions every working analyst faces: what does this dataset actually let me say, what is a fair baseline, and when does my model break down.

The deliverable specs are in the rubric. The data and starter materials are in the repo. Pick a direction, ask good questions, and let's see what you find.

Clark, C. (2026, January 30). US saw pandemic-level layoffs in 2025, and 2026 job growth may be 'uncomfortably slow.' Yahoo Finance.

<https://finance.yahoo.com/news/us-saw-pandemic-level-layoffs-140000874.html>

Harvey, J., Chou, K., & Friedman, E. (2026). Tech Employment in the Age of AI: SARIMA Forecasting of Global Tech Layoffs. DS 4002 Project 2 Repository.

<https://github.com/jensen-harvey/DS4002-project2>